Signal Without Strategy: Search Attention, Earnings Surprise, and the Difference Between Association and Tradable Prediction

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Abstract

Public information variables can be economically associated with market reactions without supplying stable incremental forecasts or implementable trading returns. This paper synthesises a connected research program that tested two such variables on United States equities: Google search attention around earnings announcements, and a vendor-defined earnings surprise for ten large semiconductor firms. We separate three distinct claims that are often conflated — association, incremental out-of-sample prediction, and tradability after timing and costs — and evaluate each against prespecified gates.

Search attention produced a standalone event strategy returning a mean net 0.8543% per event, but its incremental contribution above a strong price-and-calendar baseline was 0.0131 percentage points with p = 0.98332. Directional and magnitude models showed no incremental skill: a logistic direction model scored a worse Brier score than an intercept-only baseline (0.2711 versus 0.2560), and magnitude mean absolute error was 3.6299 with attention versus 3.6284 without (paired p = 0.9827). Earnings surprise behaved differently. It showed a genuine positive association with the initial price reaction that strengthened in an earlier temporal sample (Spearman +0.2169, p = 0.000236, over 283 out-of-sample events), yet the same frozen model improved forecast error in only five of eight years and for four of nine firms, and its bootstrap interval included zero.

We also report a leakage episode in which duplicated events and a mis-specified lag feature produced an apparent 63.6% directional accuracy that fell to 47.9% after correction. The contribution is a worked demonstration of where explanatory signal stops becoming predictive strategy.

Keywords: earnings announcements; investor attention; out-of-sample forecasting; data leakage; replication; post-earnings-announcement drift

JEL classification (proposed; to be confirmed against the target venue): G14, G17, C52, C53

Exploratory retrospective research synthesis. Not preregistered. No investable strategy is claimed.

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2. Introduction

Two ideas motivate much of the modern "alternative data" literature. The first is that public attention is informative: if unusually many people search for a company's name, something is happening to that company, and prices may respond. The second is that earnings surprise is informative: if reported profit differs from what was expected, the price should move to reflect the news. Both ideas are intuitive, both have serious academic support, and both are widely marketed as foundations for trading strategies.

This paper is about the gap between those two facts and a third claim that does not follow from them. A variable can be reliably *associated* with market outcomes without *predicting* them incrementally, and it can predict them incrementally without being *tradable* once one accounts for when information actually becomes observable and what execution costs. These are three different assertions requiring three different kinds of evidence. Conflating them is, in our reading, the most common way that plausible research becomes an unreliable strategy.

The research question. Across a connected program of retrospective studies, do search attention or vendor-defined earnings surprise supply *incremental, out-of-sample* forecasting power over strong baselines — and if so, does that power survive realistic timing and costs?

Main findings. Search attention repeatedly failed the incremental test. Strategies built around attention-selected earnings events were profitable in raw terms, but the profitability was attributable to the underlying earnings event, not to attention: once a baseline containing price and calendar information was in place, the marginal contribution of attention was economically and statistically indistinguishable from zero. Earnings surprise told a more interesting story. Its association with the first full-session price reaction was real, replicated in direction across two non-overlapping time periods, and became statistically clear in the larger, earlier sample. But the association did not translate into stable forecast-error reduction, so the downstream drift and trading stages were never authorised.

Along the way, an internal defect produced a result that looked like a discovery. Correcting it changed a directional model from apparently skilful to demonstrably unskilful. We report that episode as a finding in its own right.

Contribution. We do not claim markets are efficient, that Google Trends is useless, that earnings surprises do not matter, or that alternative data cannot work. We claim something narrower and, we think, more useful: we locate precisely where two intuitively attractive signals stop being predictive, and we document the methodological controls whose presence or absence changed the answer.

Roadmap. Section 3 defines the three-level framework. Section 4 explains the data and study architecture from first principles. Section 5 reports the search-attention results and Section 6 the leakage episode. Sections 7 and 8 cover the earnings-surprise measurement work and its outcome results including the temporal replication. Section 9 gives a cross-study evidence matrix, Section 10 discusses interpretation, Section 11 states limitations, and Section 12 concludes.

3. Conceptual framework: three levels of evidence

Level 1 — Association. A variable moves together with an outcome. Formally, some measure of dependence between the variable and the outcome differs from zero in-sample. Association is cheap to establish and is often contemporaneous: it can hold even when the variable is measured *at the same time as* or *after* the outcome begins to form.

Level 2 — Incremental prediction. The variable improves a genuine out-of-sample forecast *relative to a strong baseline*. Three words carry the weight. *Out-of-sample* means the model never saw the evaluated observation during fitting or tuning. *Incremental* means the comparison is against a baseline already containing everything cheaply available — not against an empty model. *Strong* matters because a weak baseline makes almost anything look useful; an intercept-only probabilistic forecast, for example, is a surprisingly hard benchmark to beat on calibration.

Level 3 — Tradability. The incremental forecast survives contact with implementation: the signal must be observable strictly before the position is taken, the holding period must be executable, and returns must survive transaction costs and realistic slippage.

Each level is strictly stronger than the last. Association does not imply prediction, because a contemporaneous relationship carries no guarantee that the variable leads the outcome, and because improvement against a *particular* baseline is not implied by correlation. Prediction does not imply tradability, because a statistically real reduction in forecast error can be far smaller than the cost of acting on it.

Level	Claim	Minimum evidence required	Typical failure mode
1. Association	The variable moves with the outcome	A dependence measure with an honest interval; no look-ahead in construction	Contemporaneous or post-hoc measurement
2. Incremental prediction	It improves a proper OOS forecast	Chronological validation, a strong baseline, a prespecified loss, an uncertainty interval, stability across periods	Weak baseline; in-sample tuning; leakage
3. Tradability	The forecast is implementable	Signal observable before entry; explicit entry/exit; costs deducted; clustered inference	Ignoring timing; ignoring costs; overfitting the backtest

This ladder is the organising device for everything that follows. Each study is placed at the highest level its evidence actually supports — not the highest level it was designed to reach.

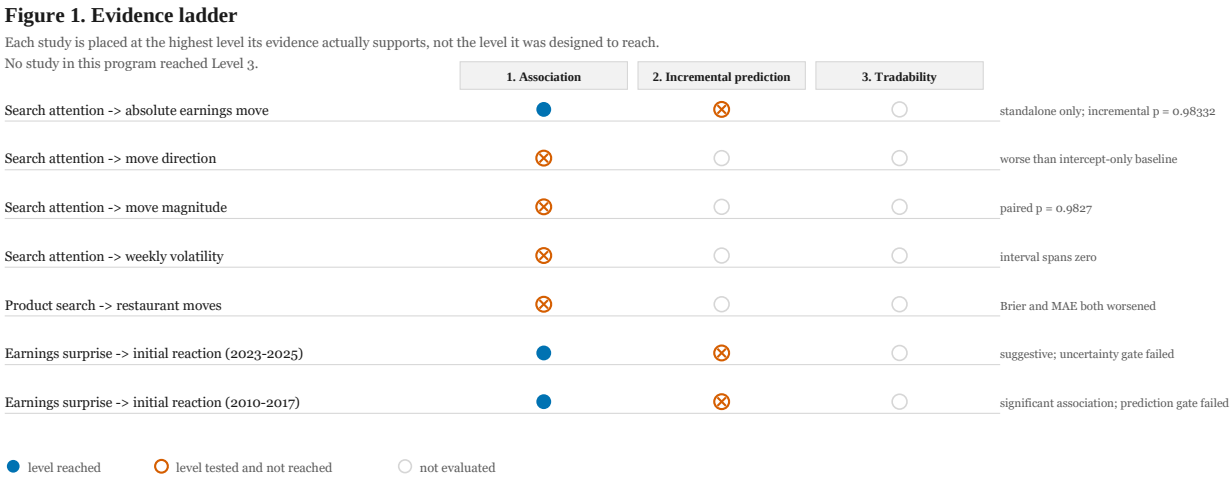


Figure 1. Evidence ladder. Each study is placed at the highest level its evidence actually supports. No study in this program reached Level 3 (tradability).

4. Data and study architecture

4.1 Google Trends and search-query ambiguity

Google Trends does not report search counts. It reports a **relative** index: for a requested query, region, and time window, Google scales the series so the maximum observed value equals 100 and everything else is expressed as a proportion of that peak. Two consequences matter. First, the index is not comparable across windows: extending the window can rescale every earlier point. Second, the series is sampled, so repeated retrievals of the identical request can differ.

A subtler problem is **query ambiguity**. A ticker such as "AMD" is also an acronym in other domains, and a company name such as "Apple" is also a fruit. Any attention series therefore measures a mixture of the intended construct and unrelated interest. We treat search attention as a *noisy proxy* for investor attention, not as a measurement of it.

Because Trends series retrieved today are a retrospective reconstruction, they cannot by themselves establish that the identical series was available at the time a decision would have been made. This limitation is recorded explicitly in the committed nowcast artifact and is not resolved anywhere in this program.

4.2 Earnings announcements and event timing

United States issuers report quarterly results via a press release, filed with the Securities and Exchange Commission (SEC) as an 8-K under **Item 2.02**, "Results of Operations and Financial Condition". The SEC records an **acceptance timestamp** — when the filing was actually received — which is the closest available proxy for when the information became public.

Timing determines which trading session can first reflect the news. We use a rule fixed before outcomes were examined, in United States Eastern Time: a release accepted before 09:30 maps to that same trading session; a release accepted at or after 16:00 maps to the next session; a release on a weekend or market holiday maps to the next session; and a release accepted *during* market hours is **excluded**, because a daily closing price cannot

separate the pre-release from the post-release part of that day. Getting this wrong is not a minor issue: mapping an after-close release to the session that has already closed would import the outcome into the predictor.

4.3 Abnormal returns and benchmarks

A raw stock return mixes firm news with whatever the market did that day. We therefore work with **abnormal returns**: the stock's return minus a benchmark's return over the identical window. For the semiconductor studies the benchmark is SMH, a semiconductor sector exchange-traded fund, which removes both market-wide and sector-wide movement and isolates the firm-specific component.

Two windows recur. The **initial abnormal reaction** spans the close before the event session to the close of the event session — the first full session in which the news can be absorbed. The **subsequent abnormal return** spans the event-session close to the close five sessions later, the window in which post-announcement drift, if present, would appear.

4.4 Vendor-defined earnings surprise, and what it is not

Earnings surprise is conventionally *reported minus expected* earnings per share (EPS). Obtaining this historically is harder than it appears, and the distinction drove a substantial part of this program.

We initially attempted to parse EPS directly from SEC earnings-release exhibits. That work is described in Section 7. When it proved infeasible at the required coverage, we moved to a commercial vendor feed. The resulting variable is labelled `VENDOR_DEFINED_REPORTED_EPS_SURPRISE` throughout, and the label is load-bearing. The vendor's `reportedEPS` is **not** SEC-validated GAAP diluted EPS: on 139 events where a vendor value overlapped an SEC-validated parsed value, only 8 matched within $0.01 — a match rate of 0.0576 — with a median absolute difference of $0.26. The differences were not random. They decomposed into retroactive share-split restatement and a GAAP-versus-adjusted definitional gap.

The estimate-vintage limitation is unresolved. The vendor publishes no guarantee that a historical `estimatedEPS` is the consensus that existed *at the announcement* rather than a value revised later. This program records that as `POINT_IN_TIME_ESTIMATE_PROVENANCE_NOT_GUARANTEED` and does not resolve it by assumption. If those estimates were revised after the fact, the surprise variable is contaminated in an unknown direction. This is the single largest threat to interpreting even our statistically clear association, and we do not claim to have eliminated it.

Because share splits scale reported EPS, estimated EPS, and their difference proportionally, we use `surprisePercentage` — a ratio — as the primary field, and verified mechanically on all 320 in-window vendor records that the ratio is invariant to proportional restatement. That neutralises *split* restatement. It does **not** neutralise *estimate-vintage* revision.

4.5 Validation, metrics, and inference

All model evaluation uses **chronological expanding-window validation**: train on all data up to a year, test on the following year, then expand. This mirrors how a forecaster would actually have operated and prevents future observations from informing past predictions. Every transformation — winsorisation limits, standardisation means and standard deviations, and hyperparameter selection — is fitted **inside the training fold only** and then applied frozen to the test fold. A single full-sample standardisation would leak distributional information from the test period backwards.

Metrics are chosen to match the target. **Mean absolute error (MAE)** measures average forecast error for continuous outcomes. The **Brier score** is the mean squared error of probability forecasts, where lower is better

and calibration matters as much as discrimination [Brier 1950]. **AUC** (area under the receiver operating characteristic curve) measures ranking ability for binary outcomes; 0.5 is chance. **Spearman correlation** measures monotonic association without assuming linearity.

For uncertainty we use a **moving-block bootstrap**: resampling calendar blocks rather than individual events preserves the fact that events cluster in earnings season and that firms within a period share shocks. Where coefficient inference is reported, standard errors are clustered by firm using a CR2/bias-reduced-linearisation estimator with Satterthwaite degrees of freedom, which is appropriate when the number of clusters is small [Cameron et al. 2008].

4.6 Leakage

Leakage is the accidental presence, in a model's inputs, of information not available at prediction time [Kaufman et al. 2012]. Two forms bit us directly. **Duplicated events** inflate apparent sample size and can place copies of the same observation in both training and test folds. **Same-event lag features** occur when a variable intended to carry the *previous* event's value actually carries the current event's own outcome. Section 6 documents both.

4.7 Status of the components

Interpretation requires knowing what was fixed in advance and what was not.

- **Frozen before designated runs.** The semiconductor extraction quality gate; the feasibility

design for the vendor data; the outcome design for the initial-reaction study; and the temporal replication design. Each was committed before the corresponding outcomes were opened.

- **Corrective controls introduced after defects were found.** Deduplication and the lag-feature fix

in the direction study (Section 6); throttle-aware retrieval in the vendor audit. These improved validity but were reactive, and we label them as such.

- **Retrospective exploratory work.** Everything in this synthesis. No result here is a

prospectively registered confirmation.

- **Never activated.** A separate prospective protocol (internally "HYP-01") was specified but never

ran, so it contributes no evidence and appears nowhere in our results.

5. Search-attention results

We report the load-bearing evidence rather than every exploratory branch, and we do not claim that all possible attention specifications have been tested.

5.1 Standalone profitability is not attention alpha

An event strategy selecting earnings events using an attention filter produced a mean net return of **0.8543%** per event. Taken alone this looks like a working signal. The decisive test is incremental: with a baseline already containing price and calendar information, the marginal contribution attributable to attention was **+0.0131 percentage points** with **p = 0.98332**. The point estimate is near zero and the p-value is close to 1.

The natural reading is that the strategy captured the well-documented tendency of prices to drift after earnings announcements [Ball and Brown 1968; Foster et al. 1984; Bernard and Thomas 1989] rather than anything contributed by search attention. Attention was, in effect, an event-selection device that picked earnings events —

and earnings events carry drift regardless of who is searching.

5.2 Direction

A logistic model predicting the sign of the initial price move was compared to an intercept-only baseline that simply assigns the training-fold base rate to every observation. On pooled out-of-sample data the baseline achieved a Brier score of **0.2560**; the logistic model achieved **0.2711**. Lower is better, so the model was *worse calibrated than assuming the base rate*. Its AUC was **0.4654**, below the 0.5 chance level.

5.3 Magnitude

A model forecasting the absolute size of the earnings move produced a pooled out-of-sample MAE of **3.6299** percentage points with attention features versus **3.6284** without them. The attention-augmented model was very slightly worse, and a paired test of the loss difference gave **p = 0.9827**.

5.4 Weekly volatility

The final weekly volatility model showed a MAE change of **–0.0309%** (baseline MAE **0.420367** versus augmented **0.420497**), with a paired block-bootstrap interval of **[–0.0011023, +0.0010839]** spanning zero.

5.5 Product search and restaurant demand

A pilot testing whether product-level search improves prediction of restaurant-sector earnings moves found that adding search blocks **worsened** both metrics: pooled Brier rose from **0.20211** to **0.23665** to **0.24739** as search blocks were added, and continuous MAE for the absolute move rose from **5.4534** to **5.6790** to **5.8543** percentage points. The committed decision label is `NO_INCREMENTAL_SEARCH_VALUE`.

5.6 Fundamental nowcasting

An attempt to nowcast company fundamentals from search demand terminated at `INSUFFICIENT_DATA`. It is reported for completeness and supports no conclusion in either direction.

Summary. Search attention reached Level 1 in places — it does track when something is happening — but did not reach Level 2 in any test we ran with a strong baseline.

6. Leakage as an empirical result

The directional study initially looked successful. With duplicates present, the logistic model scored **63.6%** accuracy, AUC **0.6395**, and Brier **0.23824**. Those numbers would ordinarily justify further development.

Two defects were then found. First, the event table contained **68** duplicated rows among **285**, leaving **217** unique event identities after deduplication. Duplicates inflate the sample and can put copies of one event on both sides of a train/test split. Second, a feature intended to carry the *prior* event's gap direction was in fact reading the **same event's own** gap direction — the model was being shown a transformation of the answer.

After deduplicating to unique identities and correcting the lag feature, the same model family scored:

- accuracy **47.9%** (below the 53.5% achieved by always predicting positive),
- AUC **0.4654**,
- direction-only strategy net return **–0.7027%**.

The pre-correction figures are **invalidated** and are reported here only to document the size of the distortion. This was an internal defect, discovered and corrected internally; we neither sensationalise it nor conceal it.

The methodological point generalises. Nothing about the pre-correction output looked anomalous: 63.6% accuracy on a binary financial target is plausible enough to be believed and modest enough not to invite suspicion. Realistic-looking alpha can be manufactured accidentally by two ordinary data-handling mistakes, which is precisely why leakage audits belong in the standard protocol rather than in the appendix [Kaufman et al. 2012].

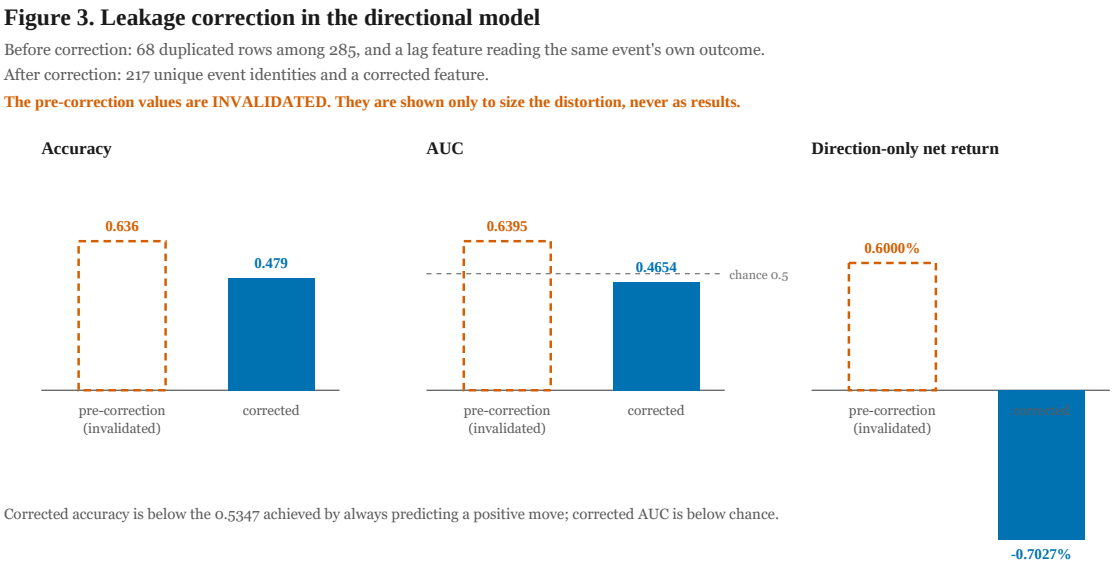


Figure 3. Leakage correction in the directional model. Pre-correction values are invalidated and are shown only to size the distortion.

7. Earnings-surprise measurement engineering

Measurement work and outcome evidence must be kept separate, because effort spent on the former is not evidence for the latter.

A general attempt to parse EPS from earnings-release exhibits across a broad universe was abandoned as infeasible. A semiconductor-specific parser performed substantially better, reaching a held-out conditional precision of **0.9860** (combined **0.9868**) — that is, where a value could be independently checked, it was almost always right.

But precision conditional on checkability is not coverage. Verified reference coverage reached only **0.7037** against a required 0.80, and usable-event coverage **0.2742** against 0.60. The binding constraint was structural rather than a parser deficiency: the audit design validated an extracted quarterly EPS against the same quarter restated in a later quarterly filing, and a **fiscal-year-end quarter is never restated that way**, because the next periodic report is the annual filing. Of **216** extractions, **47** were fiscal-year-end quarters, capping attainable coverage at **0.7824** — below the gate, independent of parser quality.

The extraction phase was therefore classified with two separate fields rather than one: a `FAILED_EXTRACTION_QUALITY_GATE` decision alongside a characterisation of `HIGH_CONDITIONAL_PRECISION_BUT_INSUFFICIENT_VERIFIED_COVERAGE`. The original single label was corrected because it implied the parser was inaccurate, which the evidence contradicts.

A subsequent audit of the vendor feed found uniform coverage (**320** complete records, 10 firms) and perfect internal arithmetic consistency, but its crosswalk gate failed at **0.6719** against 0.90 — again for a structural reason, since only **267** of **306** SEC events carried a parsed fiscal period, capping that gate at **0.8344**. That gate failed and **remains failed**; the outcome studies that follow used a successor exact-date crosswalk and are labelled accordingly.

8. Initial-reaction results and temporal replication

8.1 The 2023–2025 study

Using vendor `surprisePercentage` on **298** uniquely matched semiconductor events, with **116** out-of-sample events across 2023–2025:

	Baseline (firm effects)	**+ Surprise**
MAE of initial abnormal reaction	0.052506	**0.051203**
Improvement		**+2.482%**

Spearman correlation between transformed surprise and the initial reaction was **+0.13515** (**p = 0.148026**). The block-bootstrap 95% interval for the MAE difference was **[–0.0007272, +0.0031700]**, which includes zero. The model improved MAE in **all three** out-of-sample years.

Three of four prespecified gates passed; the uncertainty gate failed. Under the frozen rule the study stopped.

8.2 The 2010–2017 temporal replication

The terminal replication applied the identical frozen model, grid, transformation, bootstrap, and gates to an earlier, non-overlapping window, on **462** uniquely matched events with **283** out-of-sample events across 2010–2017:

	Baseline (firm effects)	**+ Surprise**
MAE of initial abnormal reaction	0.044226	**0.043246**
Improvement		**+2.216%**

Spearman was **+0.2169** with **p = 0.000236** — larger in magnitude and now clearly significant, on a sample roughly 2.4 times larger. The bootstrap interval was **[–0.0000561, +0.0021342]**, again including zero, though only marginally at the lower bound. The augmented model beat the baseline in **five of eight** years and helped **four of nine** firms — while **all nine** firms showed a positive firm-level Spearman correlation. The replication failed its prespecified gate.

8.3 Interpretation

The correct reading has four parts, and the order matters.

1. **Earnings surprise contains information about the contemporaneous reaction.** A Spearman of +0.2169 at p = 0.000236 across 283 events, positive for every one of nine firms, is not a null result and we do not describe it as one.
2. **The association replicated in direction and strengthened.** Both windows produced positive point estimates for both the MAE improvement and the correlation. The two samples do not overlap in time.

3. **The fixed model did not deliver stable forecast-error improvement.** Five of eight years, four of nine firms, and a bootstrap interval containing zero. The annual pattern was notably unstable — strong early, absent in the middle years, strong again at the end.
4. **Therefore the downstream stages were never authorised.** The information–price gap was never constructed, the drift model never fitted, and no trading was simulated. A gate that exists to prevent building on unstable foundations only works if it is honoured when it fails.

The two windows were never pooled. Combining two independent failures of the same gate to manufacture a pass would invert the purpose of the gate, and no such combination was performed.

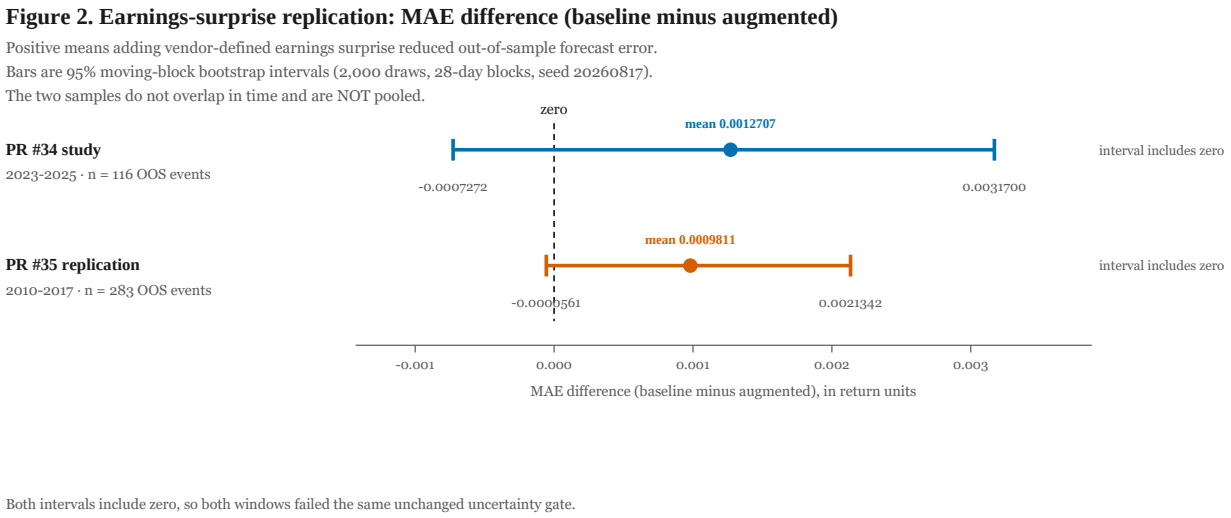


Figure 2. Earnings-surprise replication: MAE difference (baseline minus augmented), with 95% moving-block bootstrap intervals. The two samples do not overlap in time and are not pooled. Both intervals include zero.

9. Cross-study synthesis

Construct	Target	Sample	Association	Incremental prediction	Trading	Principal limitation	Final interpretation
Search attention	Absolute earnings move	Retrospective US equities	Present in places	**No** — +0.0131 pp, p = 0.98332	Standalone 0.8543% net, not incremental	Drift confound	Event selector, not alpha
Search attention	Move direction	217 unique events	None detected	**No** — Brier 0.2711 vs 0.2560; AUC 0.4654	–0.7027% net	Leakage before correction	No directional skill
Search attention	Move magnitude	Retrospective panel	Weak	**No** — MAE 3.6299 vs 3.6284, p = 0.9827	Not run	Noisy normalisation	No incremental value
Search attention	Weekly volatility	Weekly panel	Weak	**No** — –0.0309%, interval spans zero	Not run	Small effect size	No incremental value

Construct	Target	Sample	Association	Incremental prediction	Trading	Principal limitation	Final interpretation
Product search	Restaurant earnings moves	Sector pilot	Not established	**No** — Brier and MAE both worsened	Not run	Small sample	`NO_INCREMENTAL_SEARCH_VALUE`
Vendor earnings surprise	Initial reaction 2023–2025	116 OOS events	Positive, not significant (+0.13515, p = 0.148026)	**Gate failed** — +2.482%, interval spans zero	Not authorised	Estimate vintage	Suggestive, not established
Vendor earnings surprise	Initial reaction 2010–2017	283 OOS events	**Positive and significant** (+0.2169, p = 0.000236)	**Gate failed** — +2.216%, 5 of 8 years	Not authorised	Estimate vintage	Real association, unstable prediction

No new score, ranking, or aggregate is computed from this table.

10. Discussion

Why a variable can explain a reaction yet fail to forecast it. The initial-reaction studies regress a same-session outcome on a variable released just before that session. That is close to a *contemporaneous* relationship: surprise and reaction are two views of one event. Forecasting is a harder task requiring the variable to carry information the baseline lacks *and* to do so consistently enough that average error falls. A modest, unstable edge can be genuinely present and still fail to reduce mean absolute error reliably.

Rapid incorporation. If prices absorb earnings news within the first session, then by the time a daily-close-based strategy can act, the informative content is largely priced. Our design measures the reaction over the first full session and would enter only at that session's close — after the absorption we measure.

Regime and firm heterogeneity. The replication's annual pattern was strong in the early years, absent in the middle, and strong again at the end, and the model helped only four of nine firms. One natural hypothesis is that the relationship is regime-dependent. We record this as a hypothesis *generated by* the observed instability. We did not test it, it is not established, and testing it retrospectively on the same data would be exactly the multiple-testing error this program is trying to avoid [Harvey et al. 2016].

Measurement error. Both constructs are proxies. Search attention mixes investor interest with unrelated queries and is delivered on a rescaled, sampled index. Vendor EPS is restated for splits and follows an adjusted rather than GAAP definition. Classical measurement error attenuates estimated relationships, so our nulls may understate any true effect — a limitation, not a defence.

Sample size and power. With 116 and 283 out-of-sample events and firm-level clustering, our power to detect a small MAE improvement is limited. Failing an interval-based gate is at least as consistent with low power as with no effect, and we prefer that honest ambiguity to a claim in either direction.

Strong baselines. The single most consequential design choice was comparing against strong baselines. An intercept-only probability forecast beat a fitted logistic model; a firm-effects baseline absorbed most of what surprise appeared to offer. Against an empty baseline, several of these signals would have looked successful.

Why null incremental results are useful. Published financial signals are filtered by positive results, so the aggregate literature overstates how many work [Harvey et al. 2016; Bailey and López de Prado 2014]. Documented nulls with prespecified gates and honest intervals are a partial corrective — particularly when, as

here, the nulls are *incremental* nulls that leave the underlying association intact.

11. Limitations

These are material and we state them prominently.

1. **This is a set of connected exploratory studies, not one preregistered experiment.** Designs were frozen before their own designated runs, but the program as a whole evolved.
2. **This synthesis is not preregistered.** It was written after the results were known.
3. **Some controls were reactive, introduced only after defects were discovered** — above all

event-identity deduplication and the **strictly earlier lag-feature repair**. What was frozen in advance, per designated study, was the model specification, the primary metric, the decision gates and the temporal-replication rules, each committed before that study's designated outcomes were opened. That is narrower than saying the program was prespecified.

4. **Selected company universes.** Ten semiconductor firms; sector-specific pilots elsewhere. Findings need not generalise.
5. **Vendor-defined, retrospectively restated EPS**, not SEC-validated GAAP EPS.
6. **No guaranteed point-in-time estimate vintage.** The largest single threat to interpreting even the significant association.
7. **Limited cluster counts.** Nine to ten firms is few for clustered inference.
8. **Daily rather than intraday execution** in several studies; intraday releases were excluded because daily bars cannot isolate them.
9. **No historical options baseline**, so we cannot say whether options markets already priced these moves.
10. **No live capital was deployed** and no prospective result exists.
11. **We make no claim of universal market efficiency.** We tested specific signals, on specific universes, over specific windows, with specific models.

12. Conclusion

Attention and earnings surprise both carry descriptive information about how prices react to public news. In the strongest case we measured, vendor-defined earnings surprise showed a positive, statistically clear association with the first full-session reaction across 283 out-of-sample semiconductor events, positive for every firm examined.

Descriptive information alone is not alpha. The same variable, in the same frozen model, improved forecast error in only five of eight years and for four of nine firms, with a bootstrap interval including zero — and search attention, tested against strong baselines, failed to add incremental value in every specification we ran.

Method changed conclusions materially. Correcting duplicated events and one mis-specified lag feature turned an apparent 63.6% directional accuracy into 47.9%. Requiring a strong baseline turned a 0.8543% standalone strategy into an incremental contribution of 0.0131 percentage points at $p = 0.98332$. Requiring temporal replication turned a suggestive 2023–2025 result into a better-identified but still unstable 2010–2017 one.

None of the tested strategies qualifies as investable, and none is offered as such. The contribution of this work is not a signal. It is a reasonably careful map of where explanatory signal stops becoming predictive strategy — and a demonstration that the boundary sits earlier, and is easier to cross accidentally, than intuition suggests.

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Appendix A. Source artifacts

Every figure in this manuscript is drawn from a committed artifact in this repository:

- `RETROSPECTIVE_MARKET_RESPONSE_RESULTS_v1.0.json`
- `RETROSPECTIVE_DIRECTION_SIGNAL_RESULTS_v1.0.json` and
- `RETROSPECTIVE_MAGNITUDE_FORECAST_RESULTS_v1.0.json`
- `RESTAURANT_SEARCH_PILOT_RESULTS_v1.0.json`
- `SEARCH_DEMAND_NOWCAST_RESULTS_v1.0.json`
- `ATTENTION_ACTIVITY_MISMATCH_RESULTS_v1.0.json` (branch `phase2-attention-activity-mismatch`)
- `SEMICONDUCTOR_EPS_ABSORPTION_RESULTS_v1.0.json`
- `Live Signal Engine/Semiconductor_EPS_Absorption/structured_earnings_feasibility_v1.0.json`
- `SEMICONDUCTOR_SURPRISE_ABSORPTION_V3_RESULTS_v1.0.json`
- `SEMICONDUCTOR_SURPRISE_TEMPORAL_REPLICATION_RESULTS_v1.0.json`